

Vanilla LVAE on Thermoelastic 2D

Ablation study: Baseline BN → GN encoder → GN + Latent Noise

What changed, why it matters, and what the numbers say

Three Runs at a Glance

Vanilla LVAE · thermoelastic2D · seed 1 · latent_dim = 100

Metric	Baseline BN (ep 9 999)	GN encoder (ep 57 ⚠)	GN + Noise $\sigma=0.1$ (ep 1 033, running)
Encoder	BatchNorm	GroupNorm + SN	GroupNorm + SN
Latent noise σ	—	—	0.1
Early stopping	patience 10 / ep 0	patience 10 / ep 0	patience 50 / ep 2000
Train rec	0.000410	0.02939	0.008553
Val rec	0.03309	0.03246	0.03121 ✓
Gap (val/train)	80× ✗	1.10× ⚠*	3.65×
Active dims	65 / 100	100 / 100	69 / 100
Vol active?	Yes (late)	No	Yes
Status	Done	Stopped early	Running

* GN gap 1.10× is misleading — both train and val high because model hadn't converged at ep 57.

Run 1 — Baseline: BatchNorm Encoder

Full training (10 000 epochs) · the overfitting benchmark

What happened

- The model trained to completion (epoch 9 999) and memorised the training set.
- Train rec collapsed to 0.000410 — essentially zero error on training samples.
- Val rec stayed at 0.03309. The encoder learned to compress training samples perfectly but generalised only weakly.

✗ **Gap of 80x — extreme overfitting. Encoder memorised, not generalised.**

- Volume did activate (65/100 active dims) — but only because train rec was far below threshold, not because the model truly converged.

→ BatchNorm running stats cause a small eval-time distribution shift, but the dominant effect is capacity-driven memorisation.

Key numbers

Train rec	0.000410
Memorised training data	
Val rec	0.03309
Generalisation limited	
Gap	80x
Catastrophic overfitting	
Active dims	65 / 100
35 dims pruned by vol	
Volume below threshold	Yes
Activated late in training	
Epoch	9 999
Fully trained	

Run 2 — GN + SN Encoder (no noise)

Stopped at epoch 57 · aggressive early stopping prevented convergence

What happened

⚠ **patience=10 from epoch 0: most aggressive early stopping possible.**

- At epoch 57, both train rec (0.029) and val rec (0.032) are still high — model had not entered the low-loss regime.
- Early stopping triggered because val rec did not improve for 10 epochs in the very early, noisy phase of training.

⚠ **Gap of 1.10× is NOT a sign of good generalisation — both sets failed equally (underfitting).**

- Volume never activated (pruning epoch=500 not yet reached). Active dims = 100/100.

✗ **This run is not usable as an ablation. A proper GN-only run needs patience≥50 from epoch≥2000.**

Key numbers

Train rec **0.02939**

Model not converged

Val rec **0.03246**

Both high — underfitting

Gap **1.10×**

Misleading: both losses high

Active dims **100 / 100**

Pruning epoch not reached

Vol active **No**

Epoch **57**

Stopped too early

ACTION: Re-run with patience=50, start_epoch=2000, latent_noise_sigma=0.0 to get a fair ablation.

Run 3 — GN + SN Encoder + Latent Noise ($\sigma = 0.1$)

Epoch 1033 / 10 000 · still running · early stopping from epoch 2000

What's happening so far

- ✓ **Train rec = 0.008553. Learning well, but regularised — noise prevents near-zero memorisation.**
 - ✓ **Val rec = 0.03121 — already BETTER than the fully-trained baseline (0.03309).**
 - Gap = 3.65×. Still large, but massively better than baseline's 80×. Noise forces the decoder to handle imprecise codes.
Down from 80× — noise works
 - ✓ **Active dims = 69/100. Volume activated and 31 dims already pruned — at epoch 1033 vs baseline's 65 dims at epoch 9999.**
31 dims pruned at ep 1033
 - Early stopping not active yet (starts epoch 2000). ~7000 more epochs. Expect further pruning and convergence.
- Risk: if gap stays at 3.65× till ep 2000, encoder still overfits and we need stronger regularisation.

Key numbers

Train rec 0.008553

Regularised by noise

Val rec 0.03121 ✓

Better than fully-trained baseline

Gap 3.65×

Down from 80× — noise works

Active dims 69 / 100

31 dims pruned at ep 1033

Vol active Yes

Expect further pruning and convergence

Epoch 1 033 / 10 000

Still running

Analysis — Why Latent Noise Helps

Mechanistic explanation and interpretation of the gap reduction

Without noise (baseline BN)

- The encoder maps each training sample to a precise, tight latent code.
- The decoder memorises the mapping: exact code → exact design.
- For a val sample, the encoder outputs a slightly different code (distribution shift from BN) → poor fit to EM. This encoder, trained only on design codes, produces large errors.

→ **Train rec → 0, val rec stays at 0.033. Gap = 80×**

With noise $\sigma = 0.1$

- Gaussian noise is added to z before decoding. The decoder never sees the same code twice for the same sample.
- The decoder learns neighbourhood robustness: 'any code near the encoder output → reconstruct this design.'
- Train rec stays at ~ 0.009 (can't memorise noisy codes). Val rec improves because decoder handles imprecise val codes.

→ **Val rec = 0.031 (better than baseline 0.033). Gap = 3.65×**

Design choices

$\sigma = 0.1$ Conservative. 10% of typical latent std. Too large → destabilises training.

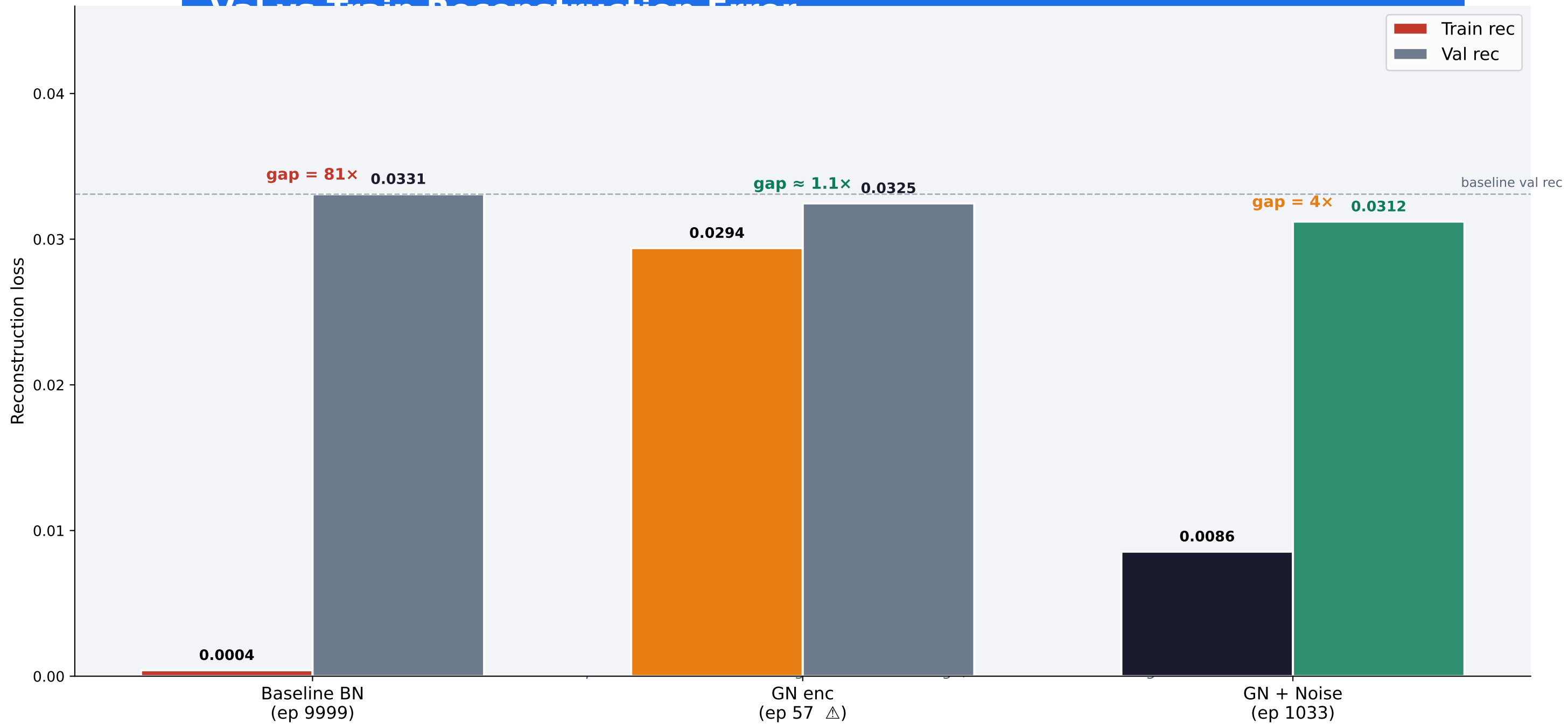
Noise only on active dims Pruned dims frozen at mean — no noise added.

Clean from BN volume improving fit to EM. This encoder, trained only on design codes, produces large errors.

Noise off at eval Val/test always use clean encoder output → fair measurement.

Analogy Dropout in the latent space. Prevents decoder from relying on exact codes.

Value Train Reconstruction Error



Projections — What to Expect When the Run Completes

GN + Noise · ep 1033 / 10 000 · early stopping from ep 2000

Likely trajectory

- Train rec will keep decreasing slowly (noise prevents zero, but model continues improving)
- Val rec may decrease further or plateau. Current 0.031 is already the best observed.
- Active dims will keep decreasing as val pressure prunes more. If val rec stays low, pruning could reach ≤ 60 dims by ep 2000
- Early stopping from ep 2000 (patience 50) will stop once val rec plateaus — the right termination criterion.

✓ **If val rec < 0.030 with ≤ 60 active dims at termination: strictly better than baseline in both quality and compactness.**

→ If gap stays at 3.65× until end: encoder still overfits. Try orthogonality regularisation + val gate combined.

Missing ablation: GN alone

GN run stopped at ep 57 — too early to be useful.

- We cannot yet separate the effect of GN encoder vs. the noise $\sigma=0.1$.

→ Proposed re-run:

- patience = 50
- latent_noise_sigma = 0.0 (no noise)

→ Fair comparison we have NOW:

Baseline BN (ep 9999, full) vs GN+Noise (ep 1033, running)

Val rec: 0.033 → 0.031 ✓

Gap: 80× → 3.65× ✓

Next Steps

In priority order

1. Wait for GN + Noise run to complete (ep 2000+)

Run is at ep 1033. Early stopping from ep 2000 determines final val rec and active dim count. Most important data point — wait before drawing conclusions.

2. Re-run GN encoder WITHOUT noise (proper early stopping)

Use `early_stopping_start_epoch=2000`, `patience=50`, `latent_noise_sigma=0.0`. Gives the missing ablation point to isolate GN vs noise effects.

3. If val gap persists ($>2\times$): test GN + Noise + Val-Gate combined

Val-gate variants are also running. If val gate alone helps, the combination (noise + val gate) may be the strongest regulariser.

4. Analyse active dim count at full convergence

Baseline: 65/100 at ep 9999. If noise run converges at <60 dims with better val rec $\rightarrow$ more compact AND generalisable — a double win.

5. Extend analysis to constrained LVAE and PLVAE

Constrained and PLVAE noise runs are also running in parallel. Check if the benefit of noise is consistent across all three architectures.